

Abdul Hayy Khan

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Profile

Motivated Python Developer specializing in backend engineering with FastAPI and Flask. Experienced in database management (SQLite, PostgreSQL) and secure system design through internships at DevelopersHub and CodeAlpha. Proven ability to deliver scalable solutions, ranging from real-time collaboration tools to self-defending vaults.

Experience

DevelopersHub Corporation, Remote

Jul 2025 – Sept 2025

Software Development Intern

- Built a Task Management System (Flask, SQLite) and an Enhanced Collaboration Platform (FastAPI, PostgreSQL).
- Integrated real-time notifications via WebSockets and secure JWT authentication for multi-user collaboration.
- Designed responsive front-end interfaces using Jinja2 templates, Bootstrap 5, and custom CSS.
- Deployed applications to cloud platforms (Railway/Render) and maintained high test coverage with Pytest.

CodeAlpha, Remote

Jul 2025 – Aug 2025

Python Programming Intern

- Completed four distinct Python projects: a Stock Portfolio Tracker, Automation Script, Chatbot, and Game.
- Utilized Regex and File Handling to build an automation tool for extracting and saving contact data from text files.
- Designed algorithms for game logic and financial tracking, ensuring accurate calculations and edge-case handling.
- Demonstrated strong grasp of Python data structures (Lists, Dictionaries) and standard libraries (Random, OS).

Education

Bachelor of Science in Artificial Intelligence

Sept 2024 – Aug 2028

Dawood University of Engineering & Technology, Karachi

Projects

Student Learning & Attendance Management System (SLAMS)

Developed a full-stack academic management platform using Flask and SQLAlchemy, featuring secure role-based access control for Admins, Teachers, and Students. The system digitizes core academic workflows, including real-time attendance tracking, course enrollment management, and secure resource sharing with approval pipelines. [Check out the more on GitHub.](#)

Enhanced Task Management System

Engineered a collaborative task platform using FastAPI and PostgreSQL, designed for high performance and scalability. Implemented real-time features like live notifications using WebSockets, secure JWT authentication, and granular permission-based task sharing. The application utilizes Pydantic for robust data validation and Jinja2 for server-side rendering. [Check out more on GitHub.](#)

Secure Vault (Active Defense System)

Developed a self-defending file management system using Python and Streamlit that integrates active security measures. Features include automated firewall blocking (via UFW) after failed login attempts, heuristic malware detection, and symmetric file encryption. The system ensures data integrity with forensic logging and SHA-256 hashing. [Check out more on GitHub.](#)

Streamlit Web Applications

Developed a suite of interactive Python web applications using Streamlit, featuring real-time API integrations and data visualization. Projects include a Healthcare Translation App (integrating speech-to-text and translation APIs), a NASA Astronomy Viewer, and various utility tools like dynamic wallpaper and quote generators. [Check out more on GitHub.](#)

Skills

Languages

- Python, SQL
- HTML5 / CSS3 (Jinja2 Templates)

Web Frameworks

- FastAPI, Flask, Streamlit

Databases

- SQLite, PostgreSQL, MySQL, Oracle

Tools & Concepts

- Git & GitHub, RESTful APIs, Authentication, Testing (Pytest, Unittest)

Core Competencies

- Problem Solving, Technical Documentation, SDLC, Debugging, Time Management